



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Complex Variables & Laplace Transform (MAT215)

Laplace Transform of Step Functions

The Second Translation Formula

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CONDUCTED BY

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What is Step Function?

The Unit Step Function

The unit step function, denoted by $u(t)$, is defined as 

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

Shifted Unit Step Function

The unit step function, denoted by $u(t - a)$, is defined as 

$$u(t - a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a. \end{cases}$$

Physical Interpretation of Unit Step Function

The second translation formula

💡 Laplace Transform of $f(t) \cdot u(t - a)$

If the Laplace transform of $f(t)$ exists, then the Laplace transform of $f(t) \cdot u(t - a)$ is given by

$$\mathcal{L}\{f(t) \cdot u(t - a)\} = e^{-as} \mathcal{L}\{f(t + a)\},$$

where $a > 0$.

Important Trigonometric Identities

Compound Angle Identities

$$\sin(A + B) = \sin A \cos B + \cos A \sin B,$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B,$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Product-to-Sum Identities

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B),$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B),$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B).$$

Negative Angle Identities

$$\sin(-A) = -\sin(A),$$

$$\cos(-A) = \cos(A).$$

How to compute $f(t + a)$?

Application of the second translation formula

② Problem

Evaluate

$$\mathcal{L}\{(2t - 3) u(t - 1)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{e^{-2t} u(t - 1)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{t^2 e^{2t} u(t-3)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{t \sin(2t) u(t - \pi)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{e^{-2t} \cos t \, u(t - 3\pi)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{te^{3t} \sin t \, u(t - \pi)\}$$



Converting a piecewise function into Step function

Laplace Transform of Piecewise Functions

② Problem

Find the Laplace transform of the function

$$f(t) = \begin{cases} 0, & 0 \leq t < 2, \\ 3, & 2 \leq t < 4, \\ e^t, & t \geq 4. \end{cases}$$

Problem

Find the Laplace transform of the function



$$f(t) = \begin{cases} 0, & 0 \leq t < \pi, \\ \cos t, & t \geq \pi. \end{cases}$$

Problem

Find the Laplace transform of the function



$$f(t) = \begin{cases} 5 \sin t, & 0 \leq t < \pi, \\ 0, & t \geq \pi. \end{cases}$$

Problem

Find the Laplace transform of the function

$$f(t) = \begin{cases} 0, & 0 \leq t < \pi, \\ \sin t, & \pi \leq t < 2\pi, \\ 0, & t \geq 2\pi. \end{cases}$$



Problem

Find the Laplace transform of the function



$$f(t) = \begin{cases} 5 \sin t, & 0 \leq t < \pi, \\ -4 \cos(2t), & t \geq \pi. \end{cases}$$

Thank You!

We'd love your questions and feedback.

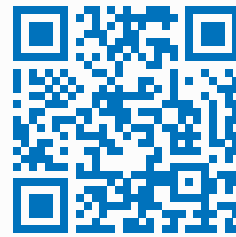
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(Lectures, walkthroughs, and course updates)



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References

- [1] Murray R. Spiegel, *Schaum's Outline of Complex Variables*, 2nd Edition, McGraw-Hill, 2009.
- [2] Dennis G. Zill, *A First Course in Differential Equations with Modeling Applications*, 11th Edition, Cengage, 2018.